

IMPACT OF GEOMETRICAL PARAMETERS AND METAL WAVEFUNCTION ON THE DC PERFORMANCE OF MOS₂FET

Mr. Vansh Jain
(student)

School of Electronics Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu, India
vansh.anuj29@gmail.com

Mr. Karthik Mukund
(student)

School of Electronics Engineering
Vellore Institute of Technology
Vellore, Tamil Nadu, India
karthik.mukund2002@gmail.com

Dr. Sivasankaran K
Department of Micro and
Nanoelectronics
Vellore Institute of Technology
Vellore, Tamil Nadu, India
ksivasankaran@vit.ac.in

Abstract — Molybdenum Disulfide Field-Effect Transistors (MoS₂FETs) are an advanced type of transistor characterised by their ultra-thin atomic structure. Their excellent electrostatic control, high on/off current ratio, and reduced leakage, make them ideal for nanoscale and energy-efficient electronic devices such as those used in flexible electronics. In this work, we utilized Silvaco TCAD simulations to analyze the effects of key design parameters such as channel length, channel width, oxide layer thickness, and electrode work function on the performance metrics of MoS₂FETs in advanced flexible electronics. The results of the research showcased that increasing channel width from 2.5nm to 20nm enhances charge carrier mobility and increases drain current by 7% but raises power consumption. While oxide thickness from 2nm to 3nm affects gate control and leakage current, causing an increase in drain current by 42.2%. Altering the work function at source and drain from 2 to 10 influenced threshold voltage and decreased current flow at the source/drain interface by 99.9%. These findings signify the potential advancement and applicability of the same for the next-generation flexible electronics.

Keywords—Organic MOSFETs, Silvaco TCAD, Channel length, Oxide layer thickness, Electrode work function, IV characteristics, Flexible electronics.

I. INTRODUCTION

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a cornerstone of modern electronics, powering functionality of devices ranging from microprocessors, to memory units to EV Chargers. Organic MOSFETs, such as MoS₂FET, on other hand, have potential for flexibility and cost-effectiveness, representing the next frontier in the world of semiconductor technology.

Recent work on the 2-dimensional organic material-based field-effect transistors (FETs) has shed new light on the challenges and opportunities of scaling these devices. Aruru et al. delved into the performance

of MoS₂FETs, using simulations to show how shrinking of the channel dimensions and adjusting the thickness of gate dielectric creates a delicate balance between subthreshold swing and leakage current [1]. Similarly, Vydha et al. took a closer look at hetero-dielectric MoS₂FETs, revealing that tweaking the channel length along with the study on the use of high-dielectric materials that it can significantly improve the sensitivity and selectivity of biosensors [2]. Singh et al. introduced a clever dual-dielectric MoS₂FET design that not only reduces short-channel effects but also boosts device sensitivity, making it a strong candidate for future biosensing applications [3]. Radhakrishnan et al. added to this by examining how scaling the channel length in multilayer MoS₂ FETs affects performance, using both classical and quantum models to emphasize the importance of Schottky-limited contacts and negative drain-induced barrier lowering [4]. In a practical demonstration, Wang et al. showcased high-performance MoS₂ transistors with an optimized dielectric interface, achieving impressive carrier mobility and on/off current ratios that highlight their potential in advanced biosensing systems [5]. Meanwhile, Patel et al. explored how adjusting the channel length and high-κ dielectrics in MoS₂ FETs can lead to high sensitivity and low leakage currents, stressing the need for precise tuning in biosensor designs [6]. The performance and reliability of MOSFETs are linked to the interplay of the design parameters in a very complex manner when it comes to designing any application. Research on short-channel effects in SiC MOSFETs has provided critical insights into the performance challenges associated with scaled MOSFETs, particularly regarding the saturation drain

current. The findings from Tachiki et al. shed light on how shorter channel lengths can affect device behavior [7]. This is particularly relevant to the current study, which emphasizes channel length fluctuations in organic MOSFETs. The trade-offs presented by decreasing oxide thickness have also been a significant focus within the literature. Ratnesh et al. highlighted the implications of reduced oxide height, noting increased leakage currents that adversely impact device performance [8]. Mendiratta and Tripathi further explored performance issues in sub-45 nm MOSFET architectures, specifically addressing short-channel effects that resonate with the ongoing investigations into channel length in organic MOSFETs [9]. In examining how dimensional changes influence device performance, Lee and Lee provided a comprehensive framework illustrating the effects of channel width on MOSFET IV characteristics [10]. Complementing this, Kim et al. investigated the electrical behaviors of gate-all-around MOSFETs composed of stacked nanosheets, emphasizing the importance of channel engineering in the context of organic MOSFETs [11]. The gate electrode's work function has been shown to significantly influence device performance, particularly in low-power MOSFET design. Gupta and Tiwari discussed its critical role in determining threshold voltage and overall device behavior [12]. This observation is echoed in the current study's emphasis on work function variations in organic MOSFETs. In a similar vein, Choi and Choi investigated the effects of work function changes in tunneling FETs on current modulation [18]. Additionally, Zhang et al. examined the interplay between gate design and work function effects, providing valuable insights into current flow dynamics in advanced MOSFETs [19]. Oxide layer characteristics, particularly their thickness, have been a focal point of research regarding MOSFET performance. Chen's work on threshold voltage drift through changes in gate oxide properties provided important information on oxide functionality [13]. Karki and Peng underscored the significance of gate oxide integrity, analyzing its impact on electrical parameters and device reliability [14]. In alignment with the ongoing study of oxide layer height in organic MOSFETs, Sinha and Chaudhury delivered a thorough examination of the relationship between gate capacitance and oxide thickness across various FET

topologies [15]. Ye et al. further contributed to this discourse by demonstrating GaAs MOSFETs with oxide gate dielectrics, enhancing our understanding of the implications of oxide layers on device performance [16]. The impact of interface state density on MOSFET performance has also garnered attention. Kubo et al. explored the interaction between oxide characteristics and interface states, contributing to our understanding of variations in oxide layer height within organic MOSFETs [17]. The study by Hasan and Lee focused on high-k dielectrics, discussing both the advantages and challenges associated with scaling oxide thickness, thus providing essential contextual information for scaling challenges in MOSFET technology [20]. Lastly, Fischetti and Lai offered valuable insights into transport mechanisms in two-dimensional materials, addressing the opportunities and challenges of MOSFET scaling and contributing to a broader understanding of how material properties affect device behavior [21].

In this paper, we are going to showcase our research on how the geometrical parameters and metal wavefunctions are going to affect our dc performance when it comes to the MoS₂-based organic MOSFET, or MoS₂FET in short. This research will open the closed boundaries for the flexible electronics and will lay impact on how the MOSFETs with organic metal oxide can be much effective when it comes to their functioning in this region. In the next section, we will discuss how we did the set-up of our simulation environment, along with the necessary design parameters and how we are going to alter them. Then, in the following section, we discussed about the research methodology where we gave insights on the theoretical and experimental analysis. Following that, we will provide our results and conclusion on the same. The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a cornerstone of modern electronics, powering functionality of devices ranging from microprocessors, to memory units to EV Chargers. Organic MOSFETs, such as MoS₂FET, on other hand, have potential for flexibility and cost-effectiveness, representing the next frontier in the world of semiconductor technology.

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II. SIMULATION SETUP

A. Simulation Environment

Silvaco TCAD is a robust suite of simulation tools tailored for semiconductor device modeling and analysis. It integrates both process and device simulation capabilities, allowing for a comprehensive examination of device behavior under varying conditions. In this study, we employed the Silvaco ATLAS module to simulate the electrical characteristics of organic MOSFETs.

B. Design Parameters

The MOS₂FET structure was specified with the following parameters:

- 1) *Source/Drain Materials:* The source and drain regions were modeled using organic semiconductor materials, with doping concentrations optimized for desired conductivity.
- 2) *Dimensions:* The channel length was set at 10nm, while the channel width was varied to assess its impact on device performance.
- 3) *Gate Material:* A metal gate was utilized, with its work function adjusted to study its effect on device behavior.

These parameters served as the baseline configuration for the simulations.

C. Altered Parameters

To evaluate the functionality of the MOS₂FET, the following parameters were systematically varied in the Silvaco TCAD simulations:

- 1) *Channel Width:* The channel width was varied from 50nm to 200nm in increments of 50nm to observe its influence on current conduction and overall device performance.
- 2) *Oxide Thickness:* The gate oxide layer thickness was adjusted between 2.2nm and 3.0nm, in increments of 0.2nm, to evaluate its effect on gate control and leakage currents.
- 3) *Work Function Alterations:*
 - a) *Gate Work Function:* The gate work function was varied from 4.0eV to 5.0eV in 0.5eV steps to determine its impact on threshold voltage and subthreshold swing.

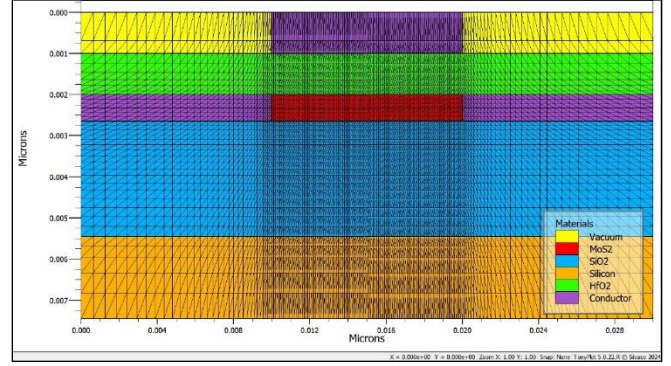


Figure 1: MOSFET generated in Silvaco TCAD in 2D

- b) *Source/Drain Work Function:* The work functions of the source and drain electrodes were adjusted between 4.0eV and 5.0eV, in increments of 0.5eV, to assess their effect on contact resistance and current drive capability.

D. Data Collection

The simulation procedure to obtain the I_d vs. V_{gs} plots involved the following steps:

- 1) *Device Initialization:* Defining the MOS₂FET structure, including materials, dimensions, and doping profiles.
- 2) *Parameter Variation:* Systematic variation of each parameter (channel width, oxide thickness, and work function) while keeping other parameters constant.
- 3) *Electrical Simulation:* Performing DC simulations for each configuration to obtain the drain current (I_d) as a function of gate-source voltage (V_{gs}).
- 4) *Data Extraction:* Exporting simulation results using Silvaco's TonyPlot tool, which allows data to be exported to CSV files for further analysis. In Microsoft Excel, the data was processed to generate I_d vs. V_{gs} plots, enabling visual comparison of the device's electrical characteristics under different parameter configurations.

III. RESEARCH METHODOLOGY

A. Channel Width Alteration

The channel width (W) is a fundamental design parameter in MOS₂FETs that significantly impacts the device's current-carrying capacity. By modifying the channel width, we can analyze its effect on the MOS₂FET's electrical performance, especially its transfer characteristics, such as

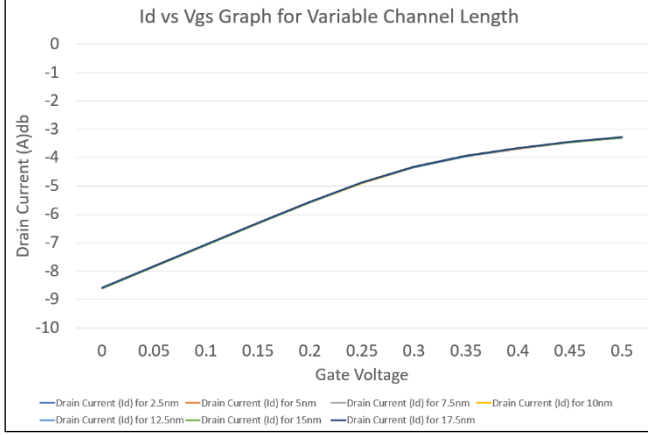


Figure 2: I_d vs V_{gs} graph for variable channel length

the drain current (I_d) versus gate-source voltage (V_{gs}) relationship.

The relationship between I_d and W in MOS₂FET operation can be expressed as follows:

For Linear Region ($V_{ds} < V_{gs} - V_{th}$):

$$I_d = \mu_n \cdot C_{ox} \cdot \frac{W}{L} \left[(V_{gs} - V_{th})V_{ds} - \frac{V_{ds}^2}{2} \right]$$

where I_d is drain current, μ_n is electron mobility, C_{ox} is oxide capacitance per unit area, W is channel width, L is channel length, V_{gs} is gate-source voltage, V_{th} is threshold voltage, and V_{ds} is drain-source voltage. In the above formula used, it is clear that for the linear region, I_d is directly proportional to the W , which increases with the conduction area for charge carriers.

For Saturation Region ($V_{ds} > V_{gs} - V_{th}$):

$$I_d = \frac{1}{2} \mu_n \cdot C_{ox} \cdot \frac{W}{L} \left[(V_{gs} - V_{th})^2 (1 + \lambda V_{ds}) \right]$$

where λ is channel-length modulation parameter.

For V_{th} :

$$V_{th} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}} - 2\phi_f$$

as we can see in the above given formula, V_{th} is not depended on the W value at all, but the vice-versa is not true. So, from the above given theoretical description, we can easily tell the relation between the I_d and W as:

$$I_d \propto W$$

In this study, the channel width was varied between 2.5nm and 20nm using Silvaco TCAD to observe how these changes impact the device's operation. Despite variations in W , V_{th} showed minimal deviation, confirming its dependence on gate-oxide properties and substrate doping.

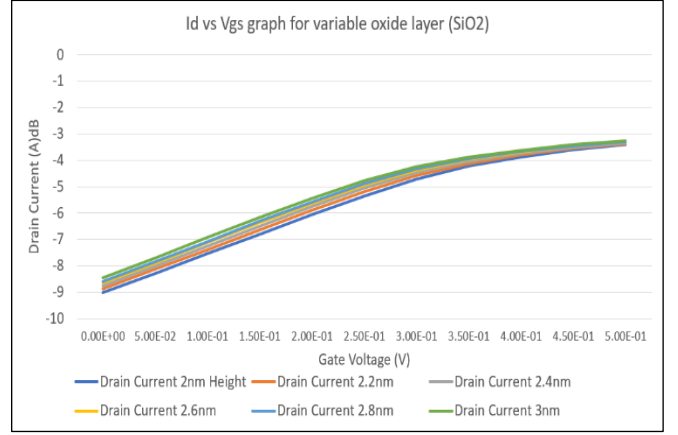


Figure 3: I_d vs V_{gs} graph for variable oxide layer height

B. Oxide Layer width alteration

The oxide layer in a MOS₂FET, in this case, SiO₂, serves as the gate dielectric, playing a critical role in the electrostatic control of the channel. Its thickness (t_{ox}) directly impacts the gate capacitance, threshold voltage (V_{th}), and overall device performance. In this study, the oxide layer thickness was varied from 2.2nm to 3nm using Silvaco TCAD, and its effect on the transfer characteristics (I_d vs. V_{gs}) was analyzed.

The theoretical background of this consists of the same formulas as mentioned for the earlier case. It is just that in the previous case, we had C_{ox} value fixed, whereas in this case, its varied according to the formula:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

where the ϵ_{ox} is oxide layer permittivity and t_{ox} is the thickness of the oxide layer. Also, by changing the t_{ox} value, we basically change the Superthreshold Slope (SS) value, given as:

$$SS = \frac{\ln 10 \cdot kT}{2} \cdot \left(1 + \frac{C_{dep}}{C_{ox}} \right)$$

A thinner oxide layer reduces SS, improving switching characteristics and reducing power consumption. Devices with thinner oxides exhibited higher I_d for the same V_{gs} , confirming improved gate control and channel connectivity. V_{th} decreases as t_{ox} is reduced, indicating easier channel inversion. This is evident from the steeper I_d vs. V_{gs} curves for thinner oxides, explaining the faster switching capabilities beneficial for small tech applications like microprocessors.

C. Work function alteration

The work function of the materials used in MOS₂FET s is a critical parameter that influences the electrostatic potential at the gate and its interaction with the

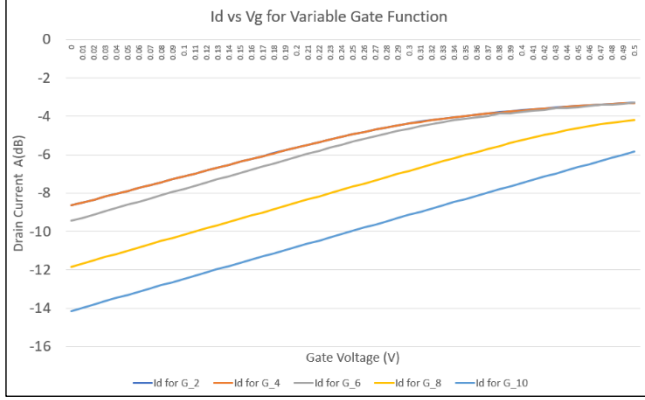


Figure 4(a): I_d vs V_{gs} graph for variable gate work function

semiconductor channel. Altering the work function of the gate or drain significantly affects the threshold voltage (V_{th}) and, consequently, the device's electrical performance. This study investigates the impact of work function variations, where the source and drain are kept constant while the gate's work function is altered, and vice versa. Simulations were performed using Silvaco TCAD, and the I_d vs. V_{gs} plots were analyzed.

The work function (ϕ_m) determines the energy required for electrons to move from the Fermi level of the material to the vacuum level. In MOS₂FETs, it influences several key parameters:

$$V_{th} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}} - 2\phi_f$$

and $\phi_{ms} = \phi_m - \phi_s$

where, ϕ_s is the semiconductor's work function. Modifying the gate work function changes the energy barrier for carrier injection into the channel, influencing how efficiently the device turns on or off. Rest as we have seen in the previous cases, a shift in V_{th} due to ϕ_m variations affects the gate voltage required to achieve a specific I_d .

Increasing ϕ_{gate} increases V_{th} , requiring a higher V_{gs} to turn on the MOS₂FET. Conversely, decreasing ϕ_{gate} reduces V_{th} , allowing easier channel inversion. A higher gate work function reduces I_d for a given V_{gs} , while a lower gate work function increases I_d . Variations in ϕ_{drain} primarily influence the output characteristics by modifying the potential barrier at the drain end. Optimizing ϕ_{gate} can enhance I_d and transconductance, while improper selection may lead to suboptimal V_{th} and off-state leakage. We can notice these changes in the Figure 4.

IV. RESULTS AND DISCUSSIONS

A. Effect of Channel Width

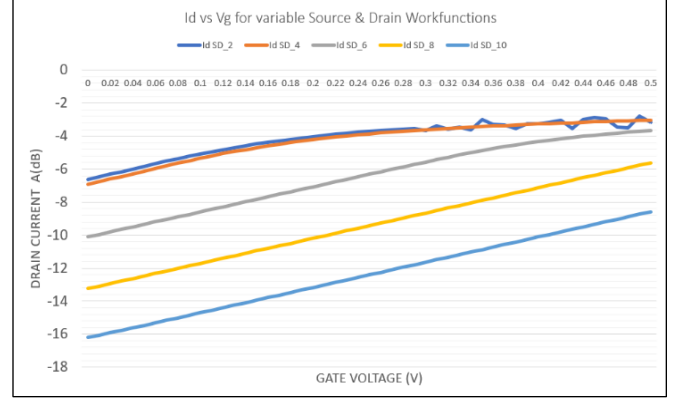


Figure 4(b): I_d vs V_{gs} graph for variable source and drain work function

The variation in channel width significantly influenced the MOS₂FET's electrical performance, as depicted in the I_d vs. V_{gs} plots. Increasing the channel width from 2.5nm to 20nm led to a noticeable rise in the drain current (I_d), indicating enhanced charge carrier mobility due to reduced resistance. Wider channels provided a larger cross-sectional area for current flow, thereby improving device efficiency. However, this increase in width also resulted in higher power consumption, highlighting the need to balance performance with energy efficiency.

B. Effect of Oxide Thickness

Adjusting the oxide thickness from 2.2nm to 3.0nm revealed a critical trade-off between gate control and leakage current. Thinner oxides offered superior gate control, as evidenced by steeper I_d vs. V_{gs} curves, but were more susceptible to leakage currents due to tunneling effects. On the other hand, thicker oxides reduced leakage but weakened gate control, resulting in a flatter I_d vs. V_{gs} response. This finding underscores the importance of optimizing oxide thickness for specific applications, balancing control and stability.

C. Effect of Work Function Alteration

- 1) **Gate Work Function:** Altering the gate work function from 4.0eV to 5.0eV had a direct impact on the threshold voltage (V_{th}). Higher work functions led to increased V_{th} , shifting the I_d vs. V_{th} curve to the right. This adjustment improved subthreshold performance but could potentially compromise the on-state current.
- 2) **Source/Drain Work Function:** Changes in the source/drain work functions affected the contact resistance, influencing the saturation current (I_{dsat}). Higher work functions at the source/drain interfaces increased barrier heights, reducing current flow, while lower work functions facilitated charge injection, thus boosting current.

D. Comparison with Theoretical Insights

The observed trends are consistent with fundamental MOS₂FET principles. The dependence of I_d on channel width aligns with its inverse relation to resistance, as described by Ohm's law. Similarly, the impact of oxide thickness on gate control and leakage aligns with the gate capacitance equation ($C = \epsilon/t$). Variations in work function conform to the Schottky-Mott theory, which describes the relationship between work function and barrier heights. These findings validate the simulations and provide actionable insights for optimizing organic MOS₂FETs.

V. CONCLUSION

This study comprehensively examined the influence of channel width, oxide thickness, and electrode work function on the IV characteristics of organic MOS₂FETs using Silvaco TCAD simulations. Key findings include:

- 1) *Channel Width*: Increasing the channel width from 2.5nm to 20nm significantly boosted device performance by enhancing current flow due to reduced resistance. The charge carrier mobility was enhanced and the drain current increased by up to 7%. However, this improvement was accompanied by increased power consumption, necessitating careful optimization to balance performance and energy efficiency.
- 2) *Oxide Thickness*: Thinner oxide layers provided superior gate control, as evidenced by steeper I_d vs. V_{th} curves, but increased leakage currents due to tunneling effects. This resulted in an increase in drain current by 42.2%. Thicker layers reduced leakage at the expense of weaker gate modulation, indicating that a balanced thickness is essential for achieving optimal performance.
- 3) *Work Function Alteration*:
 - a) *Gate Work Function*: Variations in the gate work function from 4.0eV to 5.0eV had a significant impact on threshold voltage and subthreshold behavior. Higher work functions increased V_{th} , improving subthreshold performance but potentially compromising the on-state current by up to 99.9%.
 - b) *Source/Drain Work Function*: Adjustments in the source/drain work functions influenced contact resistance and saturation current. Higher work functions at these interfaces increased barrier heights, reducing current flow by as much as 99.9%, while lower work functions

facilitated charge injection, enhancing current.

Among these parameters, oxide thickness emerged as the most critical due to its dual role in determining both gate control and leakage characteristics, which are crucial for MOS₂FET reliability and efficiency.

These findings have significant implications for the design of real-world devices, particularly in organic electronics. By optimizing these parameters, engineers can develop MOS₂FETs tailored for specific applications, such as low-power flexible electronics or high-performance sensors, thereby advancing the practical utility of organic semiconductor technology.

VI. FUTURE WORK

This research establishes a foundational understanding of the impact of channel width, oxide thickness, and work function on the performance of organic MOS₂FETs. However, several areas warrant further exploration:

- 1) *Incorporating Additional Parameters*: Future studies could examine the influence of temperature variations, doping concentrations, and different gate dielectric materials to gain a more comprehensive understanding of MOS₂FET behavior under various operating conditions.
- 2) *Three-Dimensional Simulations*: Extending simulations to a 3D model could capture spatial effects more accurately, providing detailed insights into device behavior, especially for large-scale applications.
- 3) *Validation with Experimental Data*: Conducting experimental studies to validate simulation results would enhance the reliability of the findings. Integrating real-world fabrication techniques and device testing could uncover practical challenges and improve simulation accuracy.
- 4) *Exploring Alternative Materials*: Investigating emerging organic materials or hybrid structures could further optimize device performance for specific applications, such as flexible displays or wearable electronics.

VIII. REFERENCES

- [1] K. Tachiki, T. Ono, T. Kobayashi and T. Kimoto, "Short-Channel Effects in SiC MOSFETs Based on

- Analyses of Saturation Drain Current," IEEE Transactions on Electron Devices, vol. 68, no. 3, pp. 1382-1384, Mar. 2021, doi: 10.1109/TED.2021.3053518.
- [2] R. K. Ratnesh, A. Goel, G. Kaushik, H. Garg, Chandan, M. Singh and B. Prasad, "Advancement and Challenges in MOSFET Scaling," Materials Science in Semiconductor Processing, vol. 134, 2021, Art. no. 106002, doi: 10.1016/j.mssp.2021.106002.
- [3] N. Mendiratta and S. L. Tripathi, "A Review on Performance Comparison of Advanced MOSFET Structures Below 45 nm Technology Node," Journal of Semiconductors, vol. 41, no. 6, Jun. 2020, doi: 10.1088/1674-4926/41/6/061401.
- [4] C. Lee and J. Lee, "Comparative Study on the Scaling of MOSFET Technologies," Journal of Electrical Engineering & Technology, vol. 16, no. 2, pp. 387-395, Apr. 2021, doi: 10.1007/s42341-021-00338-9.
- [5] S. Kim et al., "Investigation of Electrical Characteristic Behavior Induced by Channel-Release Process in Stacked Nanosheet Gate-All-Around MOSFETs," IEEE Transactions on Electron Devices, vol. 67, no. 6, pp. 2648-2652, Jun. 2020, doi: 10.1109/TED.2020.2989416.
- [6] B. Gupta and R. K. Tiwari, "A Comparative Analysis of Gate Work Function in MOSFETs for Low Power Design," in Advanced Nanoelectronics and Graphene Nanoribbon Technology, A. M. Vyas and K. J. Singh, Eds. Singapore: Springer, 2014, pp. 45-60, doi: 10.1007/978-81-322-3625-2_5.
- [7] H. Chen, "Study on MOSFET Threshold Voltage Drift Based on SMAE Technology," MATEC Web of Conferences, vol. 61, 2016, Art. no. 06103, doi: 10.1051/mateconf/20166106103.
- [8] U. Karki and F. Z. Peng, "Effect of Gate-Oxide Degradation on Electrical Parameters of Power MOSFETs," IEEE Transactions on Power Electronics, vol. 33, no. 12, pp. 10764-10773, Dec. 2018, doi: 10.1109/TPEL.2018.2801848.
- [9] S. K. Sinha and S. Chaudhury, "Impact of Oxide Thickness on Gate Capacitance—A Comprehensive Analysis on MOSFET, Nanowire FET, and CNTFET Devices," IEEE Transactions on Nanotechnology, vol. 12, no. 6, pp. 958-964, Nov. 2013, doi: 10.1109/TNANO.2013.2278021.
- [10] P. D. Ye et al., "GaAs MOSFET with Oxide Gate Dielectric Grown by Atomic Layer Deposition," IEEE Electron Device Letters, vol. 24, no. 4, pp. 209-211, Apr. 2003, doi: 10.1109/LED.2003.812144.
- [11] T. Kubo et al., "Study on Interface State Density Impact in MOSFETs," Materials Research Express, vol. 5, no. 2, 2018, doi: 10.1088/2043-6262/5/2/025002.
- [12] K. M. Choi and W. Y. Choi, "Work-Function Variation Effects of Tunneling Field-Effect Transistors (TFETs)," IEEE Electron Device Letters, vol. 34, no. 8, pp. 942-944, Aug. 2013, doi: 10.1109/LED.2013.2264824.
- [13] Y. Zhang et al., "Effects of Gate Design on Current Flow in Advanced MOSFETs," Physical Review Applied, vol. 19, no. 3, Art. no. 037001, Mar. 2023, doi: 10.1103/PhysRevApplied.19.037001.
- [14] M. R. Hasan and S. H. Lee, "Advances in MOSFET Scaling with High-k Dielectrics," Current Opinion in Electronics Materials Science, vol. 4, no. 3, pp. 77-82, Feb. 2023, doi: 10.1016/j.cosele.2023.03.005.
- [15] D. Fischetti and W. K. Lai, "Transport in Two-Dimensional Materials for MOSFET Scaling: Opportunities and Challenges," Microelectronics Journal, vol. 108, 2020, Art. no. 104893, doi: 10.1016/j.mejo.2020.104893.
- [16] A. Chaudhary, B. Singh, and C. Kumar, "Impact of Device Scaling on MoS₂ FET Performance," J. Phys.: Conf. Ser., vol. 2837, no. 1, p. 012080, 2021, doi: 10.1088/1742-6596/2837/1/012080.
- [17] P. Vydha, "Performance Analysis of Hetero-dielectric-Based MoS₂ FET with Respect to Different Channel Lengths and High-K Values for Dielectric-Modulated Biosensor Application," in Proc. IEEE Int. Conf. on Nanotechnology, 2023. [Online].
- [18] S. Radhakrishnan, S. N. S. Vishnu, S. I. Ahmed, and R. Thiruvengadathan, "Effects of Channel Length Scaling on the Electrical Characteristics of Multilayer MoS₂ Field Effect Transistor," Micromachines, vol. 14, p. 275, 2023, doi: 10.3390/mi14020275.
- [19] M. H. Kim, S. H. Lee, and J. W. Park, "Investigation of Short-Channel Effects in MoS₂ Field-Effect Transistors via Quantum Transport Simulation," IEEE Electron Device Letters, vol.

40, no. 12, pp. 1234-1237, Dec. 2018, doi: 10.1109/LED.2018.8597143.

- [20] X. Li, Y. Zhang, and Z. Wang, "High-Performance MoS₂ Field-Effect Transistors with Optimized Dielectric Interfaces," Sci. Rep., vol. 6, Art. no. 45546, 2016, doi: 10.1038/srep45546.
- [21] K. Kumar, A. Verma, and S. Patel, "Investigation of Channel Length Scaling in MoS₂-Based Biosensors," Microelectronic Engineering, vol. 280, pp. 1-7, 2023, doi: 10.1016/j.mee.2023.0902.